

Abstract

Purpose: A centralized platform for managing, analyzing and visualizing financial data.

Problem: Finance teams rely on scattered spreadsheets and manual processing, which causes calculation errors, duplicated work and delayed reports.

Solution: Fin-Cash imports CSV/XLSX data, generates income statements, balance sheets and cash flow reports, shows interactive KPI dashboards, forecasts trends and protects data with role-based access control (RBAC).

Conclusion: Requirements and the Design Thinking study are complete; the project is moving to prototyping, development and testing.

Introduction

Purpose: Replace manual spreadsheet work with one web platform for financial reporting, dashboards and forecasting, for finance staff, analysts, managers/executives and administrators.

Problem: Data is fragmented across spreadsheets, reports take long to prepare, and managers have no visual summary of key KPIs.

Prior solutions: Teams process scattered spreadsheets by hand, which is slow and error-prone.

Our solution:

- **Strengths:** automated reports, real-time KPI dashboards, forecasting, RBAC, audit log, encryption (TLS and AES-256), export to PDF/XLSX/CSV.
- **Weaknesses:** does not make banking transactions or replace accounting systems; needs clean, structured data; legacy database integration is a future enhancement.

Reference: Design Thinking (Stanford d.school) and ISO/IEC/IEEE 29148:2018.

Materials

- CSV and XLSX sample financial datasets
- Modern web browsers: Chrome, Edge, Firefox
- UI/UX wireframes and mockups (dashboards, analyst workflow)
- Surveys, interviews and observation notes (Empathize stage)
- SRS document based on ISO/IEC/IEEE 29148:2018
- Test plan and test cases file
- GitHub repository for the Fin-Cash source code

Results and data analysis

User research (Empathize): surveys, interviews and observations with finance teams and executives showed three core problems: data fragmentation, slow report generation and no visual KPI summary.

Targets set for the system: dashboards load within 3 seconds, 99.5% uptime in business hours, daily automatic backup, and zero calculation discrepancy against source data.

Validation plan: usability, functional, security/RBAC, performance and data accuracy testing.

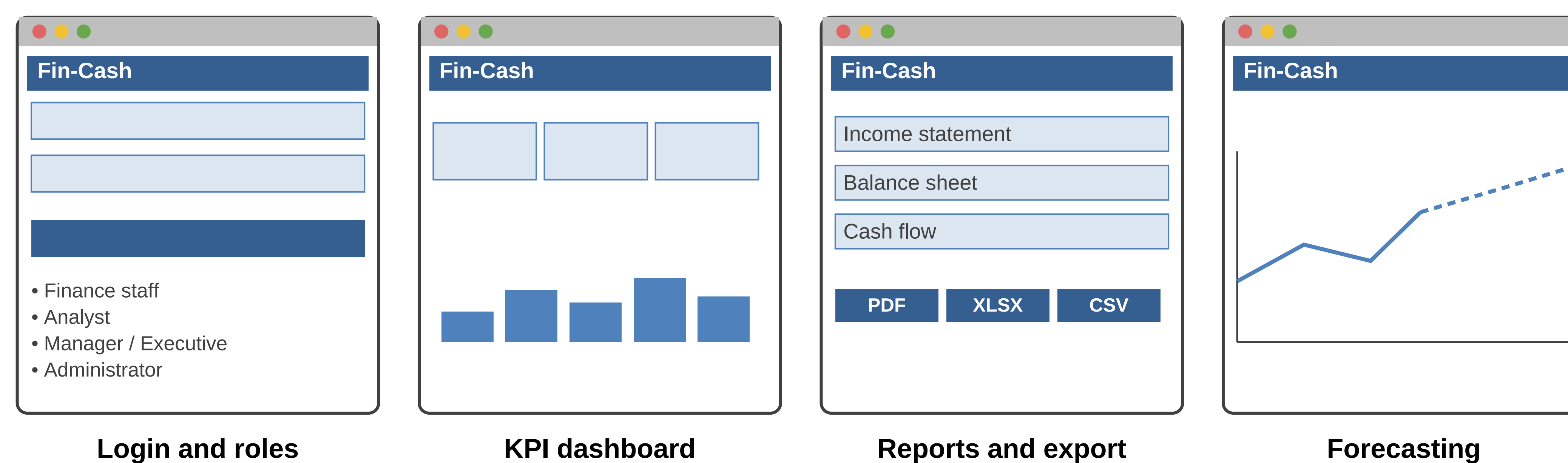
Survey graphs (pie / bar) and their explanation: add your survey results here.

Survey Link: (add your Google Form link)

Business Model Canvas

Key Partners • Corporate IT teams (provide data) • Finance departments • Academic supervisors	Key Activities • Import and validate data • Automate reports • Forecast trends	Value Proposition • One central platform • Less manual work and fewer errors • Faster reports • KPI dashboards for quick decisions	Customer Relationship • User training (minimal) • Feedback and testing sessions	Customer Segments • Finance staff • Financial analysts • Managers and executives • Administrators
	Key Resources • Financial datasets • Web platform and database • Project team		Channels • Web browser access • Intranet or internet	
Cost Structure • Development and testing time (semester timeline) • Hosting, security and backup			Revenue Streams • Subscriptions: monthly or yearly plans per organization • Extra fees for added users or advanced forecasting (possible)	

Prototype



Illustrative wireframes of the planned screens

Conclusions

Finished yet? Not yet. The project is in progress.

Why? Prototype, development and testing phases are still ahead.

Achievements:

- SRS completed: 8 functional and 6 non-functional requirements
- Design Thinking stages completed and user pain points identified
- UI/UX mockups designed for executive dashboards and analyst workflows
- Test plan prepared

Recommendation

- Integrate with corporate legacy databases (future enhancement)
- Test with real users before final deployment
- Define roles and access rights before deployment

Methods

- Empathize: surveys, interviews and observations with finance teams and executives
- Define: data fragmentation, slow reports, no visual KPI summary
- Ideate: automated reporting, forecasting and interactive dashboards
- Prototype: UI/UX mockups for dashboards and analyst workflows
- Test: user feedback to improve usability
- Wrote the SRS following ISO/IEC/IEEE 29148:2018
- Next: database and RBAC setup, data import, report engine, testing

References

ISO/IEC/IEEE. (2018). *Systems and software engineering: Life cycle processes: Requirements engineering* (ISO/IEC/IEEE 29148:2018).

Stanford d.school. (n.d.). *Design thinking guidelines*. Stanford University.